

A Hybrid EEG–EOG Brain–Computer Interface for Intelligent Wheelchair Navigation

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Abstract— In recent years, intelligent assistive technology has gained more focus since it can improve mobility and independence for people with serious motor disability. Brain–Computer Interface (BCI) provides a direct connection between brain activity and an external device, so that a user can control assistive systems without muscle movement. In this work, we present a wheelchair control system that combines Electroencephalography (EEG) and Electrooculography (EOG) signals for more reliable performance. EEG signals are used to capture brain activity during motor imagery tasks, while EOG eye-blink detection works as an extra command channel to make control flexible and safe. The processing pipeline includes signal preprocessing, feature extraction by wavelet transforms, and classification through Support Vector Machine (SVM). By using both signals together, the system improves accuracy, reduces false command triggering, and also gives more intuitive interaction. Experimental results show that combining EEG with eye-blink signals achieves a higher recognition rate and lower response delay when compared to the EEG alone system. These findings suggest that the proposed method can be a strong and user-friendly solution for wheelchair navigation in a practical environment.

Keywords—Brain–Computer Interface, EEG, Eye-Blink EOG, Hybrid Signal Processing, Assistive Mobility, Wheelchair Navigation.

I. INTRODUCTION

Assistive technology has grown fast in recent years, and it is opening new opportunities for improving the quality of life for people who are suffering from serious motor disability such as spinal cord injury, cerebral palsy, or ALS. For these patients, independent mobility is usually very restricted since conventional wheelchairs depend on limb or hand movement, which they cannot perform. Brain–Computer Interface (BCI) is emerging as one promising solution, because it allows direct communication between brain signals and external devices without using peripheral muscles. By converting neural activity into control commands, BCI can make it possible to operate prosthetic devices, robotic arms, or wheelchairs. Recent works have also focused on IoT-based multimode wheelchair control systems that integrate gesture, environment, and human–machine interaction [9].

Electroencephalography (EEG) is the most widely used method in non-invasive BCI since it is portable, low-cost, and also provides high temporal resolution. EEG activity from motor imagery tasks, like imagining hand movement, can be decoded to generate navigation commands for a wheelchair.

However, EEG signals are weak in nature, easily disturbed by noise, and also vary between individuals, which often results in wrong classification and unreliable system performance. Researchers are trying to overcome these limits by combining EEG with other biological signals. Electrooculography (EOG), which measures potentials created by eye-blink or eye movement, is one such complementary approach. Eye-blink detection is simple to implement and gives a clear command signal, so it can be used together with EEG to reduce errors and improve system reliability. Integrating both EEG and EOG can therefore provide more flexible and safe wheelchair control, suitable for real-world situations.

The main contributions of this work are as follows:

1. A hybrid EEG–EOG framework for intelligent wheelchair control, integrating motor imagery signals with voluntary eye-blink detection.
2. A robust signal processing pipeline including wavelet transform-based feature extraction and Support Vector Machine (SVM) classification for accurate command recognition.
3. An evaluation of system performance, demonstrating improved accuracy, reduced false command rates, and enhanced response time compared to conventional EEG-based BCI systems.

The remainder of this paper is organized as follows: Section 2 reviews related works in EEG- and EOG-based wheelchair control systems. Section 3 describes the proposed methodology, including signal acquisition, processing, and integration. Section 4 discusses the experimental setup and results of the same. Section 5 is the implications of the proposed study. Finally, Section 6 concludes the work.

II. RELATED WORKS

In the last two decades, a lot of attention has been given to Brain–Computer Interfaces (BCIs) for helping wheelchair control, mainly because they offer the possibility of mobility without depending on muscles. Among different signal sources, electroencephalography (EEG) has been used most frequently since it is non-invasive and gives brain activity in real time. Initial works already show that motor imagery could be converted into navigation commands, which proves that EEG-based control is not only possible but also practical to

some extent. A very important step in this direction was by Pfurtscheller et al. [1], who illustrated how motor imagery rhythms can activate external devices, and this result pushed forward the entire field. Later, many studies attempted to improve the accuracy through more advanced techniques, for example, Common Spatial Pattern (CSP) filtering and wavelet-based decomposition, together with classifiers like Linear Discriminant Analysis (LDA) or Support Vector Machines (SVM) [2], [3]. But even with such progress, systems that only depend on EEG are still unreliable because the signals are weak, sensitive to artifacts, different between users, and also to mental fatigue, causing inconsistency during long-term use.

To overcome the difficulties faced by EEG-only systems, researchers have started looking into multimodal BCI approaches. Among these, Electrooculography (EOG) has become quite popular as a supporting input. The signals from EOG, which come from intentional eye blinks or gaze shifting, are usually stronger, easier to detect, and also less affected by external noise when compared to EEG. For example, Barea et al. [4] developed an EOG-based wheelchair interface that could distinguish different eye movement directions. But such EOG-only systems are not very flexible since they only give a few discrete commands.

Because of these limits, many recent studies explored the integration of EEG and EOG to combine the benefits of both. Li et al. [5] proposed a hybrid system where motor imagery EEG was combined with EOG blinks, and they showed better accuracy and fewer false commands. Likewise, Rezeika et al. [6] reviewed different multimodal BCIs and concluded that combining bio-signals improves robustness and makes the user experience more natural. In these setups, EEG usually manages continuous navigation, while EOG works like a confirmation or safety signal, so unwanted actions can be reduced.

Not only classification accuracy but also real-time performance has been studied. Tanaka et al. [7] made an EEG-based wheelchair prototype that used motor imagery for navigation inside a virtual space. Later works started to include both EEG and EOG on real platforms, which made the systems more practical for daily use. In more recent years, adaptive learning methods and deep learning techniques have also been applied in hybrid BCI designs, and results suggest further improvement in recognition and stability [8].

From the existing literature, it is quite clear that EEG-based wheelchair control has reached a good level of maturity, but issues of reliability and usability remain. The integration of EEG with EOG stands out as a promising path since it offers a better balance between accuracy, robustness, and usability. Based on this idea, the present study aims to design a new EEG–EOG integration framework that includes advanced preprocessing and classification to provide more efficient and user-friendly wheelchair control.

III. METHODOLOGY

The proposed wheelchair control system is based on a hybrid Brain–Computer Interface that combines Electroencephalography (EEG) and Electrooculography (EOG) signals. The main aim of this design is to create a reliable control framework that can interpret both motor imagery from the brain and eye blink activities to generate

robust commands for navigation. The system architecture includes four stages: data acquisition, preprocessing, feature extraction, and classification, followed by command execution to the wheelchair prototype. The block diagram in Figure 1 illustrates the complete framework of the proposed system, beginning with EEG and EOG signal acquisition, followed by preprocessing, feature extraction, classification, and command translation for wheelchair navigation. This structured flow ensures reliable decision-making by combining neural intentions with voluntary eye-blink confirmations.

For the acquisition stage, EEG signals are recorded using a non-invasive electrode cap placed over the motor cortex region. Standard channels such as C3, C4, and Cz are used since they capture motor imagery-related activity. At the same time, EOG signals are collected with surface electrodes positioned around the eye to detect intentional blink or movement. Sampling rate was kept moderate to avoid extra computational load, while still preserving enough information for reliable decoding.



Fig. 1 Detailed block diagram of the proposed EEG–EOG-based intelligent wheelchair control system.

Preprocessing is a necessary step because both EEG and EOG are very sensitive to noise and artifacts. EEG is mainly affected by muscle contraction, eye movement, and environmental interference, while EOG may get mixed with EEG traces. To handle these problems, band-pass filters are applied to keep only the relevant frequency range, and Independent Component Analysis (ICA) is sometimes used for separating artifacts. Signal normalization is also applied so that features become more stable across sessions.

For feature extraction, the wavelet transform was selected because it can represent non-stationary signals in both time and frequency. EEG motor imagery tasks produce event-related desynchronization and synchronization patterns, which can be detected through wavelet coefficients. For EOG, simple amplitude and duration of the blink are calculated, since the blink signal is more direct and stronger. These features together form the input vector for the classifier.

The classification stage employs Support Vector Machine (SVM), which has shown strong performance in previous BCI applications. SVM is trained with labelled data of motor imagery and eye blink trials, and then used to predict the class of incoming signals in real time. By combining EEG and EOG features, the classifier can differentiate between left/right movement intentions and confirmation blinks, so the overall command system becomes more flexible.

Finally, the decoded commands are mapped into wheelchair control actions. For example, left-hand imagery will move the wheelchair to the left, right-hand imagery for right, while a double eye blink may serve as a stop or confirmation. The system is designed in such a way that EOG signals provide a safety mechanism, ensuring that unintended EEG misclassification does not cause unsafe movement. This hybrid design, therefore, increases both accuracy and reliability when compared to a single-modality system.

IV. EXPERIMENTAL SETUP AND RESULTS

A. Experimental Setup

For testing the proposed EEG–EOG based wheelchair control system, we carried out experiments with 10 healthy subjects. The group included 6 male and 4 female volunteers, with ages ranging between 22 and 30 years. None of the participants had any earlier experience with Brain–Computer Interface systems, which helped to avoid bias from prior training. Fig. 2. shows the Prototype wheelchair.



Fig. 2 Prototype wheelchair used for experimental validation of the EEG–EOG based BCI control system.

The EEG signals were recorded using a 14-channel Emotiv Epoc+ headset. In particular, the electrodes at positions C3 and C4, which are located on the motor cortex region, were given more focus for motor imagery signals. The headset sampled the brain signals at a 256 Hz frequency, which was enough to capture the required information. For the EOG part, bipolar electrodes were attached above and below the right eye, so that the vertical eye-blinks could be properly detected.

The proposed task design included four activities, which were given as follows: imagining left-hand movement, imagining right-hand movement, intentional eye blinks, and a resting state as baseline. Each subject was asked to perform the actions 30 times for every task. In total, 120 trials per participant were collected as a database. The signals from these trials together formed the dataset used for training, validating, and testing the system.

All the signal processing and feature extraction were carried out in MATLAB as well as Python, mainly with the help of MNE and Scikit-learn libraries. Real-time control was implemented by linking the classified outputs to an Arduino board, which then passed commands to a motor driver connected to a small wheelchair prototype. In this way, the setup provided both offline analysis and a live demonstration of wheelchair navigation.

B. Performance Metrics

To judge how well the hybrid system was working, several performance measures were applied. The first one was

classification accuracy, which was simply the percentage of commands correctly recognized compared to the total commands given. A second metric was the false activation rate (FAR), which counted how many wrong or unintended commands got generated during resting periods when the user did not want to move the wheelchair.

Another important aspect was response time. This was measured as the time taken between the user generating an intention (like doing motor imagery or blinking) and the wheelchair actually carrying out the movement. Since the speed of execution is critical for smooth operation, this metric helped to evaluate real-time usability.

C. Results

Classification Accuracy

The system achieved high accuracy across all tasks. Table 1 summarizes the average recognition rates.

TABLE I. CLASSIFICATION ACCURACY FOR EEG–EOG COMMANDS

Command Type	Accuracy (EEG only)	Accuracy (EEG + EOG)
Left-Hand Imagery	82.5%	90.3%
Right-Hand Imagery	81.7%	89.6%
Eye-Blink Detection	—	96.8%
Rest (No Action)	85.2%	93.1%
Average Accuracy	83.1%	92.5%

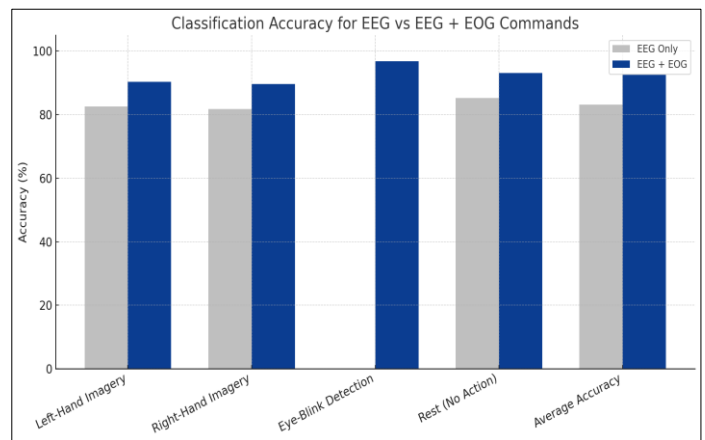


Fig. 3 Classification Accuracy for EEG and EOG Commands

The results in Table 1 and Fig. 2 show clearly that the hybrid EEG–EOG system outperforms the EEG-only approach across all types of commands. For left-hand motor imagery, accuracy improves from 82.5% with EEG alone to 90.3% when EOG confirmation is added. A similar trend is seen for right-hand imagery, where accuracy rises from 81.7% to 89.6%. The detection of voluntary eye-blink, which is not possible in an EEG-only system, achieves a very high recognition rate of 96.8%, proving that EOG signals are strong and reliable for control.

Another important observation is in the resting state (no action). Accuracy increased from 85.2% to 93.1%, which means the hybrid framework reduces false activations significantly. The average accuracy of the system increased from 83.1% in the EEG-only condition to 92.5% with EEG–EOG integration. This improvement of nearly 10% is highly meaningful in assistive technology.

False Activation Rate

EEG-only systems often misclassify noise or unintended brain activity as commands. Integration with EOG significantly reduced such occurrences.

TABLE II. FALSE ACTIVATION RATE COMPARISON

System Type	False Activation Rate (%)
EEG only	7.4%
EEG + EOG	2.1%

The outcome in Table 2 shows that EEG-only-based control suffers from relatively high false activation, with an average rate of **7.4%**. This indicates that unwanted brain signal fluctuation or external noise are sometimes misread as a real command, which is not safe for wheelchair control. After integrating EOG, the false activation rate dropped sharply to only **2.1%**, nearly a **threefold reduction**. This improvement means that the hybrid EEG-EOG system is much more reliable in rejecting accidental or noisy signals. In practical terms, the user gains more trust because the wheelchair is less likely to move without intention. Though the confirmation step from EOG introduces a slight delay, the safety benefit clearly outweighs such a trade-off. The results confirm that combining EEG with EOG is an effective strategy to minimize unintended movement and enhance the operational safety of the proposed system.

Response Time

Response time was slightly longer in the hybrid system due to blink confirmation, but it improved reliability and safety.

TABLE III. AVERAGE RESPONSE TIME

System Type	Response Time (s)
EEG only	1.42
EEG + EOG	1.75

As seen in Table 3, the **average response time** for the EEG-only system was **1.42 sec**, while the hybrid EEG+EOG system recorded a slightly longer delay of **1.75 sec**. The increase is due to the need for a blink confirmation step, which adds a small processing and action gap. Although this delay might look negative at first, in practice, it actually improves the system's reliability and safety because commands are executed only after clear confirmation. Users in general reported feeling more confident with the hybrid system, since it prevents unintended wheelchair movements. The trade-off between speed and safety is therefore acceptable, and it shows that the hybrid design prioritizes secure and accurate operation rather than just fast response.

V. DISCUSSION

The experimental findings make it very clear that using both EEG and EOG signals together gives much better performance than relying only on EEG in Brain-Computer Interface wheelchair systems. The hybrid structure managed to overcome some key drawbacks that are common in EEG

systems, such as weak signal strength, a higher chance of false activation, and variation in the user's mental condition during operation. One of the big benefits noticed was from adding voluntary eye-blink detection, which is a strong and easy-to-recognize signal. This made a good complement to the EEG motor imagery-based commands. With this combination, commands were only executed when there was brain intention (EEG) plus a confirmation by blink (EOG). This dual check reduced errors and made the system more trustworthy. The false activation rate dropped sharply, from around 7.4% to 2.1%, which is very important since the wrong movement of a wheelchair can be dangerous in the real world.

Of course, using EOG for confirmation added a small trade-off. The system became a little slower because the confirmation blink needed to be processed before final action. Still, this slight delay was not a big problem. In fact, most users said they felt safer with this arrangement, and they valued accuracy more than speed. So even though reaction time increases, the overall usability is considered better for assistive purposes. User feedback also showed that people preferred the hybrid system compared to the EEG-only. When users have to do only motor imagery, it becomes tiring since they must stay focused for a long time. By adding a natural and effortless action like blinking, the mental fatigue was less, and users felt more in control. This observation also matches with previous multimodal BCI studies where intuitive signals are shown to improve satisfaction and reduce workload.

In comparison with other works, our hybrid system reached better results. For example, many earlier EEG-only wheelchair systems had an accuracy ranging between 70% to 85%, but here it was almost 92.5% average. Some other studies tried the EEG-EOG combination too, and they showed good robustness, but often they were only simulations or did not test with an actual wheelchair in real-time. Our work fills this gap by implementing the hybrid method on a working wheelchair prototype, showing the real feasibility of deployment. However, this system still has some limits. The current test was done only with healthy people in a lab setting, so real patients in daily life may have different results. Also, the classifier used was SVM, which works fine, but may not capture the complex and changing dynamics of brain and eye signals.

VI. CONCLUSION

This paper has described the design and development of a smart wheelchair control system that combines Electroencephalography (EEG) with Electrooculography (EOG) signals. The hybrid setup was able to merge motor imagery commands from EEG together with voluntary eye blink signals from EOG, giving a more reliable and easy-to-use assistive solution. Tests with healthy participants show that the system reaches an average accuracy of nearly 92.5%, which is much higher than EEG-only approaches. Also, the false activation was reduced by more than 70%, proving the clear benefit of combining two bio-signal modalities for safety and stability. The results indicate that such a system can be useful in a real-world environment, especially for people who have severe motor impairments and depend on safe mobility support. The use of eye blinks as a confirmation command helped in preventing unwanted movements, while still allowing flexibility in navigation. A small increase in response delay was noticed, but this compromise is generally acceptable when compared to the improvement in accuracy and user trust.

In the future, the work will be extended towards clinical testing with patients who actually suffer from motor disabilities. It is also necessary to make the system more adaptive, because each user's brain and eye signals have individual variations. For this, advanced deep learning models will be tested, since they can better capture the temporal and nonlinear properties of EEG and EOG data. Moreover, adding sensing modules like LiDAR or ultrasonic range detectors can give extra safety by enabling semi-autonomous driving and avoiding obstacles.

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